ALAN GUTIERREZ RAMIREZ

C++ SOFTWARE DEVELOPER | GRAPHICS & REAL-TIME SYSTEMS

Valencia, Spain · +34 603 586 727 · alangutier24@gmail.com
Portfolio · GitHub · LinkedIn

PROFESSIONAL SUMMARY

C++ Software Developer with professional experience developing real-time physics simulation software and engineering tools for industrial training systems. Experienced in C++, NVIDIA PhysX, OpenGL and SQL, with additional hands-on work building custom graphics software and gameplay systems. Interested in software engineering, simulation, graphics programming and game engine development.

TECHNICAL SKILLS

- **Languages:** C++, C#, Python, SQL, GLSL, HLSL
- **Graphics & Simulation:** OpenGL, NVIDIA PhysX, real-time rendering, physics simulation
- **Game Development:** Unreal Engine 5, Unity
- **Software Engineering:** OOP, memory management, multithreading, debugging, performance optimization
- **Tools:** Visual Studio, CMake, Git, RenderDoc, Perforce

PROFESSIONAL EXPERIENCE

Junior Software Developer — LSyM, Laboratorio de Simulación y Modelado
Valencia, Spain · November 2024 – October 2025

- Developed a C++ tool for creating training tests and scenarios for a heavy-machinery simulation system, supporting instructors in preparing and managing training sessions.
- Contributed to the simulator's core systems and physics functionality using C++ and NVIDIA PhysX.
- Investigated and debugged issues across simulation behaviour, physics models and engine functionality, supporting QA and system reliability.
- Worked with SQL databases and translated technical documentation and manuals from Spanish into English.

SELECTED PROJECTS

SmallEngine — Personal C++ Graphics Project
GitHub

- Building a lightweight graphics engine using C++, OpenGL 4.1 and GLSL, with window, input, camera, mesh, material and rendering systems.
- Implemented volumetric rendering techniques including ray marching, Beer–Lambert transmittance, volumetric lighting and procedural density generation.
- Developed a multithreaded CPU reference renderer to validate GPU shader output; applied RAII and move semantics to GPU resource management.

Shelley Manor — 3D Narrative Game
Unreal Engine 5 · C++ · Blueprints · Released on Steam

- Contributed to the development of a 3D narrative game, implementing gameplay systems and camera controls.
- Worked with lighting, materials and post-processing, and helped debug and optimize visual and gameplay functionality.
- Project received recognition at Latam Games Com 2025 and the Game Development World Championship (GDWC).

EDUCATION

HND in Video Game Programming
ESAT — Escuela Superior de Arte y Tecnología · Valencia, Spain · July 2024

Computer Science Coursework
Gustavus Adolphus College · USA · 2014–2017

ADDITIONAL INFORMATION

- **Languages:** Spanish (Native), English (Fluent)
- **Work authorization:** European Union citizen
- **Location & availability:** Valencia, Spain · Open to remote, hybrid and on-site opportunities across Europe